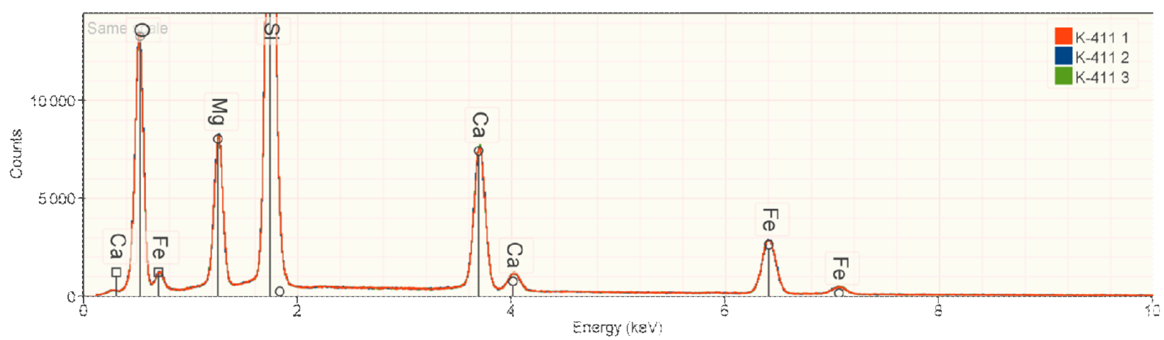
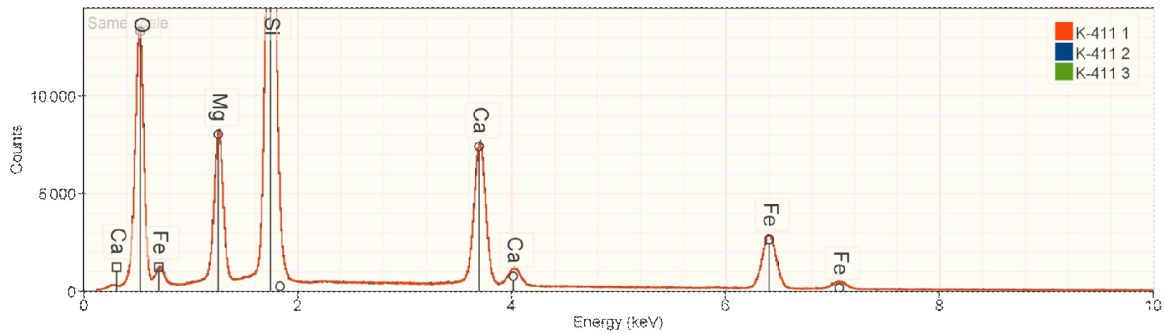
Quantification

A brief look at how quantification of x-ray spectra is performed

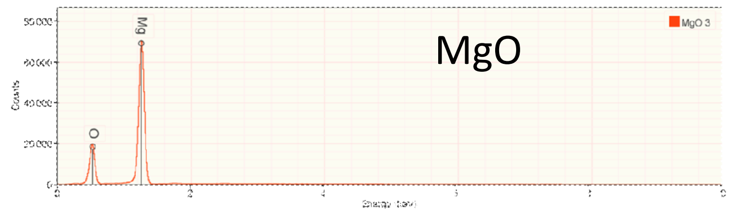
What material produces a spectrum that looks like this?




K-411 1
K-411 2
K-411 3
Counts
10 000
5 000
0
Energy (keV)
O
Mg
Si
Ca
Fe

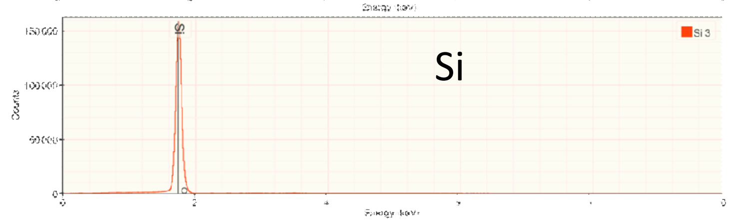
=

?


MgO
MgO 3
Mg
O

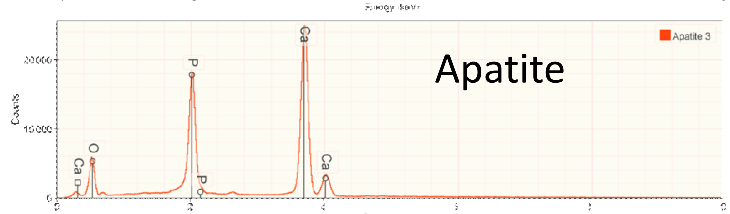
+

?


Si
Si 3

+

?


Apatite
Apatite 3
Ca
P
O

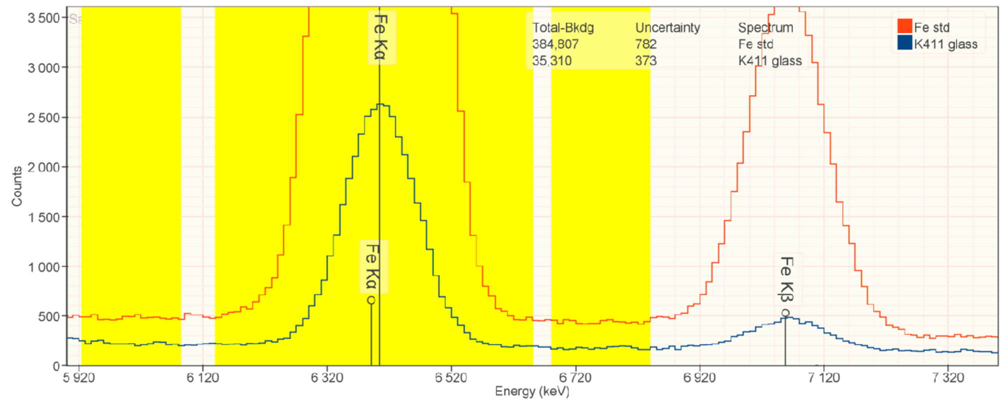
+

?

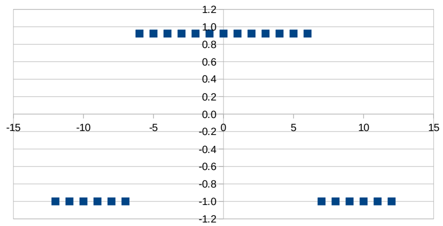

Fe
Fe 3

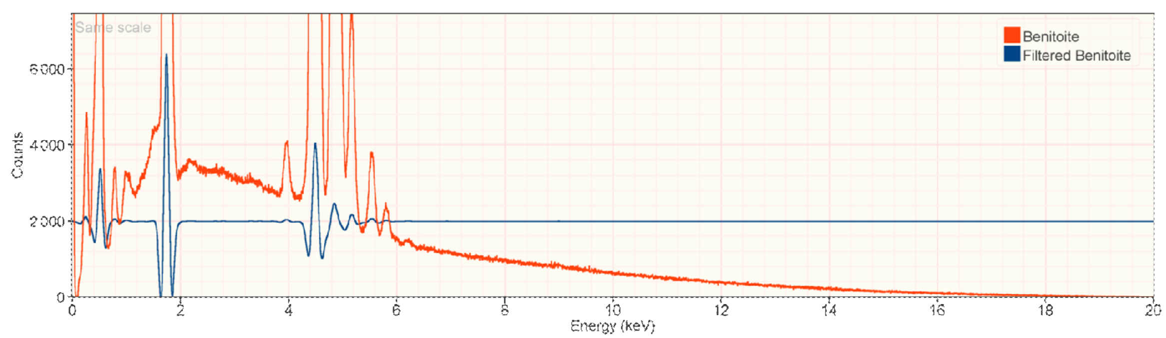
+

k-ratios

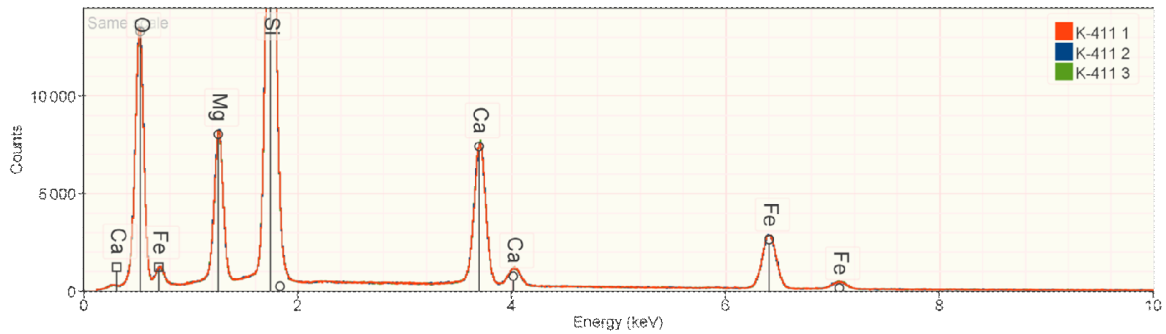


$$k = \frac{35{,}310 \pm 373}{384{,}807 \pm 782} = 0.0917 \pm 0.0010$$



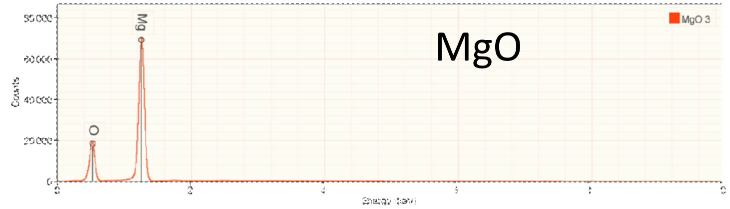


k-ratios



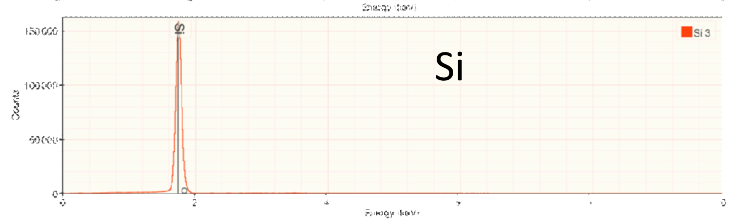
=

0.1076



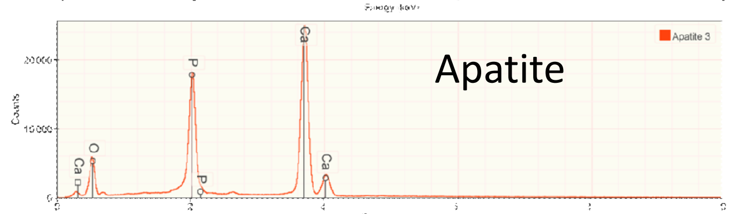
+

0.1792



+

0.2947

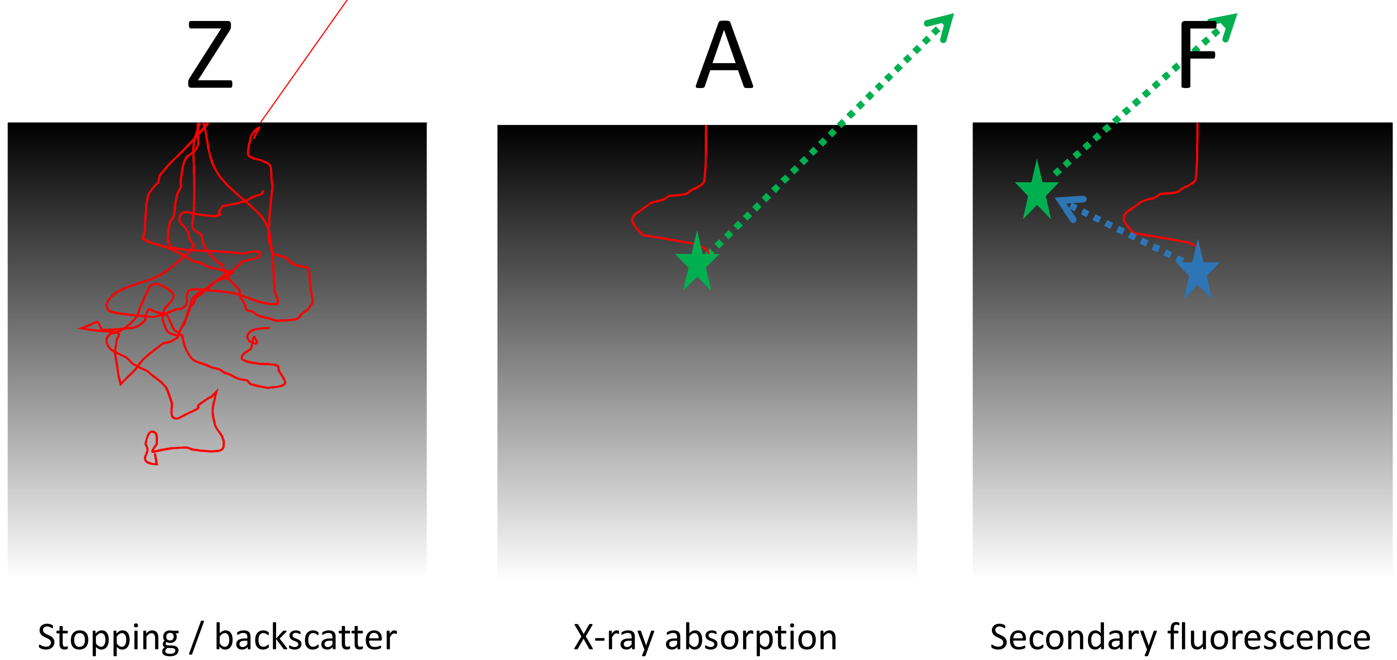


+

0.0921

+

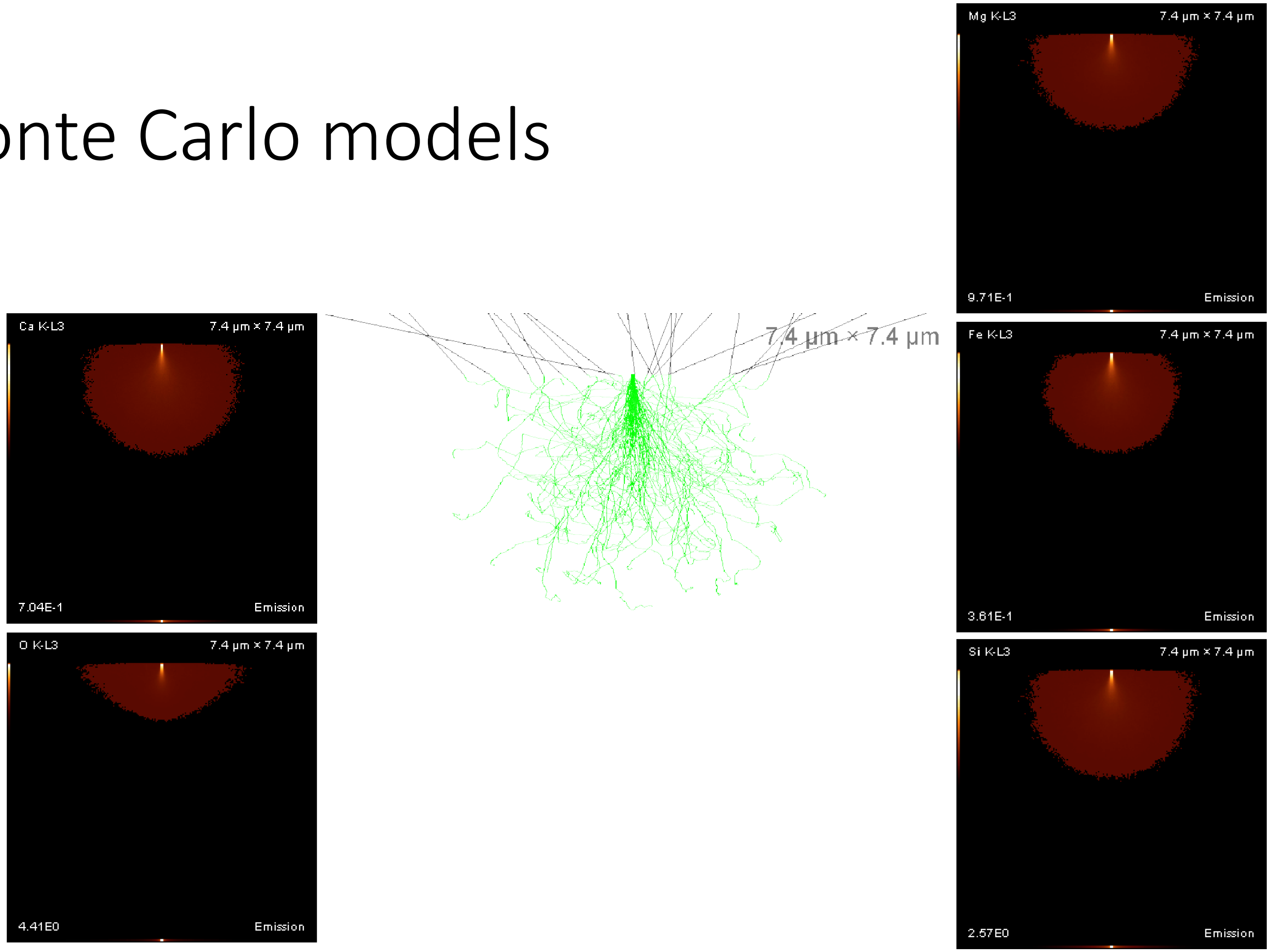
Matrix correction



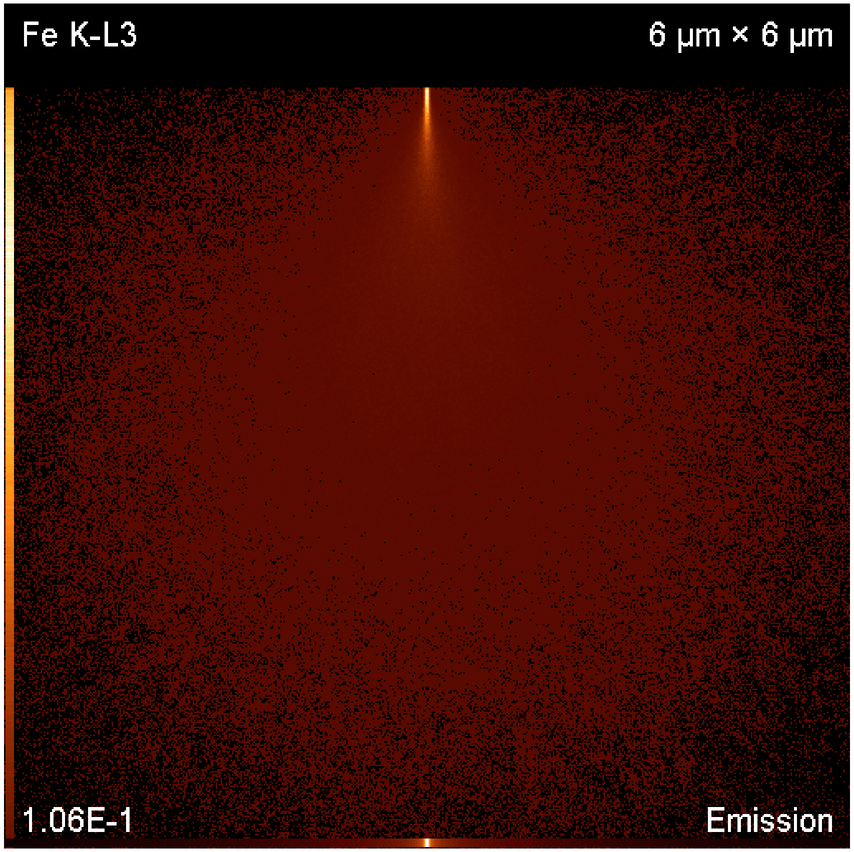
$$C_A = \frac{\{Z \cdot A \cdot F\}_{Unk,A}}{\{Z \cdot A \cdot F\}_{Std,A}} \, k_A$$

Matrix corrections turn k-ratios into compositions

Monte Carlo models

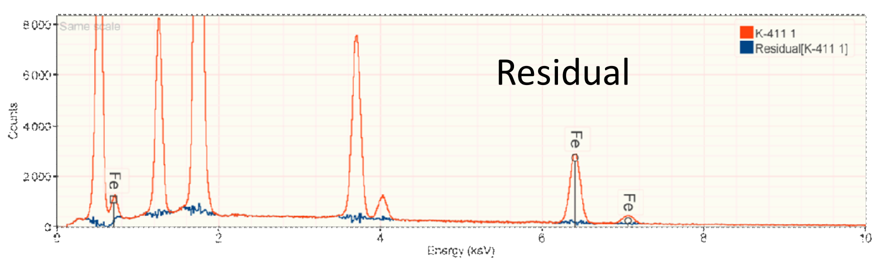


Matrix correction



Results

Spectrum	Quantity	O			Mg			Si			Ca			Fe			Sum
K-411 3 Bulk	Line				Mg All			Si All			Ca K-family			Fe Kα			
	Z · A · F	-	-	-	1.024	0.738	1.007	0.982	0.719	1.002	0.986	1.000	1.006	0.842	0.991	1.000	
	k-ratios	-	-	-	0.1076	±	0.0004	0.1792	±	0.0004	0.2947	±	0.0013	0.0921	±	0.0007	
D-H = 20.123 keV	mass fraction	0.4222	±	$8.5 \cdot 10^{-5}$(I[std,Ca]) $4.9 \cdot 10^{-5}$(I[std,Fe]) $7.0 \cdot 10^{-5}$(I[std,Mg]) 0.0002(I[std,Si]) 0.0002(I[unk,Ca]) 0.0002(I[unk,Fe]) 0.0002(I[unk,Mg]) 0.0005(I[unk,Si]) 0.0003([μ/ρ]) $7.7 \cdot 10^{-6}$(η) [0.0007]	0.0853	±	0.0001(I[std,Mg]) 0.0003(I[unk,Mg]) $7.2 \cdot 10^{-5}$([μ/ρ]) $1.1 \cdot 10^{-6}$(η) [0.0003]	0.2535	±	0.0002(I[std,Si]) 0.0005(I[unk,Si]) 0.0002([μ/ρ]) $6.0 \cdot 10^{-6}$(η) [0.0005]	0.1144	±	0.0002(I[std,Ca]) 0.0004(I[unk,Ca]) $1.2 \cdot 10^{-5}$([μ/ρ]) $2.4 \cdot 10^{-7}$(η) [0.0005]	0.1103	±	0.0002(I[std,Fe]) 0.0008(I[unk,Fe]) $4.8 \cdot 10^{-6}$([μ/ρ]) $8.1 \cdot 10^{-8}$(η) [0.0008]	0.9858
I = 12.600 nA	norm(mass fraction)	0.4283	±	0.0003	0.0866	±	0.0003	0.2571	±	0.0004	0.1161	±	0.0005	0.1119	±	0.0007	-
LT = 300.0 s	atomic fraction	0.6031	±	0.0005	0.0802	±	0.0002	0.2062	±	0.0001	0.0653	±	0.0002	0.0451	±	0.0003	



Element	Standard	Measured	Difference	% Difference
Si	25.38	25.35	-0.03	-0.1
Fe	11.21	11.03	-0.18	-1.6
O	42.37	42.22	-0.14	-0.3
Ca	11.06	11.44	0.39	3.5
Mg	8.85	8.53	-0.31	-3.5
5	98.86	98.58	0.55	5.3

Assumptions

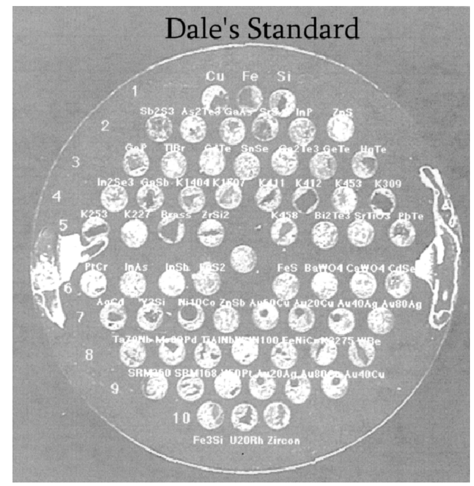
- Homogeneous material
- Flat
- Polished
- Clean
- Mounted normal to the electron beam
- Conductive or conductively coated
- Standards / unknown measured under same conditions

What did I gloss over?

- How were the samples prepared?
- Why did I select the standards I selected?
- Why did I select the beam energy I selected?
- Are any references required?
- Which line (K, L or M) should I use to quantify?
- How many x-rays do I need to collect (count time / probe dose)?
- Where do I get standard materials?

Geller

SPI



Custom

Why is it that we are willing to spend $50k+ on a detector but not an additional $10k on accessories to make our detector more accurate?